



1 determine the required function for matrix A:

$$a) \quad A = \begin{bmatrix} -3 & -8 & -9 \\ 5 & 11 & 9 \\ -1 & -2 & 1 \end{bmatrix} \quad f(A) = 4\sqrt{2} - 1$$

$$A = \begin{bmatrix} -3 & -8 & -9 \\ 5 & 11 & 9 \\ -1 & -2 & 1 \end{bmatrix} \quad \det(A - \lambda I) = 0 \quad \begin{vmatrix} -3-\lambda & -8 & -9 \\ 5 & 11-\lambda & 9 \\ -1 & -2 & 1-\lambda \end{vmatrix} = 0$$

$$-3-\lambda((\lambda-11)(\lambda-1)+18) + 8(5(1-\lambda)+9) - 9(-10+11-\lambda) = 0 \quad \lambda^3 - 9\lambda^2 + 24\lambda - 16 = 0$$

$$(\lambda-1)^2(\lambda-4) = 0 \quad \lambda_1 = 1, \lambda_2 = \lambda_3 = 4$$

$$\text{For } \lambda = 1: (A - \lambda I)v_1 = 0 \quad \begin{bmatrix} -4 & -8 & -9 \\ 5 & 10 & 9 \\ -1 & -2 & 1 \end{bmatrix} v_1 = 0 \quad \begin{bmatrix} -4 & -8 & -9 \\ 0 & 0 & -9 \\ 0 & 0 & -13 \end{bmatrix} v_1 = 0 \quad x_3 = 0, -4x_1 - 8x_2 = 0$$

$$x_1 = -2x_2, x_2 = t$$

$$v_1 = [-2t \ t \ 0]^T \quad v_1 = [-2 \ 1 \ 0]^T$$

$$\text{For } \lambda = 4: (A - \lambda I)v_2 = 0 \quad \begin{bmatrix} -7 & -8 & -9 \\ 5 & 7 & 9 \\ -1 & -2 & -3 \end{bmatrix} v_2 = 0 \quad \begin{bmatrix} -7 & -8 & -9 \\ 0 & 9 & 18 \\ 0 & 0 & 0 \end{bmatrix} v_2 = 0 \quad x_1 = x_3 = -\frac{1}{2}x_2$$

$$v_2 = [t \ -2t \ t]^T \quad v_2 = [1 \ -2 \ 1]^T$$

$$20 \quad (A - \lambda I)v_3 = v_2 \quad \begin{bmatrix} -7 & -8 & -9 \\ 5 & 7 & 9 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -7 & -8 & -9 & 1 \\ 5 & 7 & 9 & 2 \\ -1 & -2 & -3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -7 & -8 & -9 & 1 \\ 0 & 9 & 18 & 9 \\ 0 & 6 & 12 & -6 \end{bmatrix} \quad \begin{bmatrix} -7 & -8 & -9 & 1 \\ 0 & 9 & 18 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad x_3 = t, x_2 = -1-2t, x_1 = 1+t$$

$$25 \quad v = [1+t, -1-2t, t]^T$$

$$v_3 = [1 \ -1 \ 0]^T \quad P = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad J = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 1 \\ -1 & -2 & 3 \end{bmatrix}$$

$$f(A) = 4\sqrt{2} - 1 \Rightarrow f'(A) = \frac{2}{\sqrt{2}}$$

$$f(A) = P \begin{bmatrix} f(\lambda_1) & 0 & 0 \\ 0 & f(\lambda_2) & 0 \\ 0 & 0 & f(\lambda_3) \end{bmatrix} P^{-1} = P \begin{bmatrix} f(\lambda_1) & 0 & 0 \\ 0 & f(\lambda_2) & f'(\lambda_2) \\ 0 & f(\lambda_3) & f'(\lambda_3) \end{bmatrix} P^{-1}$$

$$= \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 7 & 1 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} -2 & -10 & -11 \\ 6 & 15 & 10 \\ -1 & -2 & 4 \end{bmatrix}$$

b) $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ $f(A) = A^3 - 3A^2 + 2I$ Cayley-Hamilton Theorem

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = -\lambda^3 + 5\lambda^2 - 7\lambda + 3 = 0$$

$$A^3 - 5A^2 + 7A - 3I = 0$$

$$A^3 - 3A^2 + 2I = A^3 - 5A^2 + 7A - 3I + 2A^2 - 5A + 5I = 0$$

$$A^3 - 3A^2 + 2I = A^3 - 5A^2 + 7A - 3I + 2A^2 - 5A + 5I = 0$$

$$A^3 - 3A^2 + 2I = A^3 - 5A^2 + 7A - 3I + 2A^2 - 5A + 5I = 0$$

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$$A^3 - 3A^2 + 2I = A^3 - 5A^2 + 7A - 3I + 2A^2 - 5A + 5I = 0$$

$$A^3 - 3A^2 + 2I = A^3 - 5A^2 + 7A - 3I + 2A^2 - 5A + 5I = 0$$

$$\lambda = 1 \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow \begin{cases} x_1 + x_2 + x_3 = 0 \\ x_3 = t, x_2 = -t \\ x_1 = -x_2 - x_3 = -(-t) - t = 0 \end{cases}$$

$$v = [-x_1 \ t \ t]^T \Rightarrow v_1 = [-1, 1, 0]^T, v_2 = [-1, 0, 1]^T$$

$$P = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}, P^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\Rightarrow f(A) = 4862A^2 - 9727A + 4865I$$

$$A^2 = P D^2 P^{-1} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$\Rightarrow f(A) = \begin{bmatrix} 24310 & 19448 & 19448 \\ 0 & 4862 & 0 \\ 19448 & 19448 & 24310 \end{bmatrix} + \begin{bmatrix} -19454 & -9727 & 9727 \\ 0 & -9727 & 0 \\ -9727 & -9727 & -19454 \end{bmatrix} + \begin{bmatrix} 4865 & 0 & 0 \\ 0 & 4865 & 0 \\ 0 & 0 & 4865 \end{bmatrix}$$

$$= \begin{bmatrix} 9721 & 9721 & 9721 \\ 0 & -4865 & 0 \\ 9721 & 9721 & 9721 \end{bmatrix}$$

$$c) A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad f(A) = e^A$$

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 & 0 & 0 \\ 0 & 2-\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 1 \\ 0 & 0 & 0 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda_{1,2} = 2, \lambda_{3,4} = 0$$

$$\lambda = 2 \Rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \quad v_1 = 0 \Rightarrow x_1 = t, x_2 = 0, x_3 = 0, x_4 = 0$$

$$v_1 = [t \ 0 \ 0 \ 0]^T \Rightarrow v_1 = [1 \ 0 \ 0 \ 0]^T$$

$$(A - \lambda I)v_2 = v_1 \Rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} v_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow x_2 = 1, x_1 = x_2 = x_3 = 0$$

$$v_2 = [0 \ 1 \ 0 \ 0]^T$$

$$\lambda = 0 \Rightarrow \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad v_3 = 0 \Rightarrow x_1 = x_2 = x_4 = 0, x_3 = r$$

$$v_3 = [0 \ 0 \ r \ 0]^T \Rightarrow v_3 = [0 \ 0 \ 1 \ 0]^T$$

$$(A - \lambda I)v_4 = v_3 \Rightarrow \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} v_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow x_4 = 1, x_1 = x_2 = x_3 = 0$$

$$v_4 = [0 \ 0 \ 0 \ 1]^T$$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I, \quad P^{-1} = I, \quad D = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$f(A) = e^A, \quad f'(A) = e^A$$

$$f(A) = P^{-1} f(D) P = f(D) = \begin{bmatrix} e^2 & e^2 & 0 & 0 \\ 0 & e^2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 $A_{n \times n} \rightarrow$ eigen values: $\{\lambda_1, \lambda_2, \dots, \lambda_n\} \Rightarrow f(A) \rightarrow$ eigen values: $\{f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)\}$

for diagonalizable A we can have $A = PDP^{-1}$, $D = \text{Diag}(\lambda_1, \dots, \lambda_n)$

Diagonal values of D are A 's eigen values.

5 then we know: $f(A) = P f(D) P^{-1}$, which $f(D) = \begin{pmatrix} f(\lambda_1) & & \\ & f(\lambda_2) & \\ & & \ddots \\ & & & f(\lambda_n) \end{pmatrix}$

$f(D) = \text{diag}(f(\lambda_1), \dots, f(\lambda_n))$

now, $f(D)$ is diagonal and its eigen values are diagonal entities, means: $f(\lambda_1), \dots, f(\lambda_n)$

and now, $f(D)$ and $f(A)$ are similar and similar matrices have same eigen values.

10 proof: $Ax = \lambda x \Rightarrow P^{-1}Ax = \lambda P^{-1}x \Rightarrow \underbrace{P^{-1}A}_{B} \underbrace{(PP^{-1})x}_{y} = \underbrace{\lambda}_{\lambda} \underbrace{P^{-1}x}_{y} \Rightarrow By = \lambda y : \lambda \text{ is same}$

Therefore $f(A)$ eigen values are: $f(\lambda_1), \dots, f(\lambda_n) \Rightarrow f(A)v = f(\lambda)v$

3

15 For $A = \begin{bmatrix} \alpha & \beta & \gamma \\ 0 & \alpha & \beta \\ 0 & 0 & \alpha \end{bmatrix} = \alpha I + \beta N + \gamma N^2$ then: $f(A) = f(\alpha)I + \beta f'(\alpha)N + (\gamma f'(\alpha) + \frac{1}{2!} \beta^2 f''(\alpha))N^2$

Taylor series: $f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\alpha)}{k!} (z-\alpha)^k \Rightarrow f(A) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\alpha)}{k!} (A-\alpha I)^k$

20 $A = \alpha I + \beta N + \gamma N^2 \Rightarrow A - \alpha I = \beta N + \gamma N^2$

$A - \alpha I = \beta N + \gamma N^2 \Rightarrow (A - \alpha I)^2 = (\beta N + \gamma N^2)^2 = \beta^2 N^2 + 2\beta\gamma N^3 + \gamma^2 N^4 \stackrel{N^3=0}{=} \beta^2 N^2$

$\Rightarrow (A - \alpha I)^k = 0, k \geq 3$

$f(A) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\alpha)}{k!} (A - \alpha I)^k = \frac{f^{(0)}(\alpha)}{0!} I + \frac{f^{(1)}(\alpha)}{1!} (A - \alpha I) + \frac{f^{(2)}(\alpha)}{2!} (A - \alpha I)^2 + \sum_{k \geq 3} \frac{f^{(k)}(\alpha)}{k!} (A - \alpha I)^k$

25 $\Rightarrow f(A) = f(\alpha)I + f'(\alpha)(\beta N + \gamma N^2) + \frac{f''(\alpha)}{2}(\beta^2 N^2) = f(\alpha)I + f'(\alpha)\beta N + f'(\alpha)\gamma N^2 + \frac{f''(\alpha)}{2!}\beta^2 N^2$
 $\Rightarrow f(A) = f(\alpha)I + \beta f'(\alpha)N + (\gamma f'(\alpha) + \frac{\beta^2}{2!} f''(\alpha))N^2$

4 For $A_{n \times n}$ (diagonalize) $\Rightarrow P = [x_1 | x_2 | \dots | x_n]$, eigen vectors. $\Rightarrow u' = Au, u(0) = c$
 $\Rightarrow u = \xi_1 e^{\lambda_1 t} x_1 + \xi_2 e^{\lambda_2 t} x_2 + \dots + \xi_n e^{\lambda_n t} x_n, P\xi = c$

$$A_{n \times n} = PDP^{-1}, D = \text{Diag}(\lambda_1, \dots, \lambda_n)$$

as $P = [x_1 | x_2 | \dots | x_n]$ is independent, we can write vector as combination:

$$u = [x_1 | x_2 | \dots | x_n] [y_1 | y_2 | \dots | y_n]^T = y_1 x_1 + y_2 x_2 + \dots + y_n x_n \Rightarrow u = P y \quad (u(t) = P y(t))$$

P is invertible: $y(t) = P^{-1} u(t)$ write as function

$$\Rightarrow u(t) = P y(t) \Rightarrow u'(t) = P y'(t) \xrightarrow{u' = Au} u'(t) = A u(t) = A P y(t) \Rightarrow P y'(t) = A P y(t)^*$$

$$\Rightarrow A = P D P^{-1} \Rightarrow A P = P D^* \Rightarrow P y'(t) = P D y(t) \Rightarrow P^{-1} P y'(t) = P^{-1} P D y(t)$$

$$\Rightarrow y'(t) = D y(t) \quad y(t) = (y_1(t), \dots, y_n(t))^T \Rightarrow y_i'(t) = \lambda_i y_i(t), i = 1, \dots, n$$

diagonal entries of D

can consider solution as: $y = y_0 e^{\lambda t} \Rightarrow y_i(t) = \xi_i e^{\lambda_i t}$

$$\text{Then: } y(t) = [\xi_1 e^{\lambda_1 t}, \dots, \xi_n e^{\lambda_n t}]^T$$

$$\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

$$\text{we had: } u(t) = P y(t) \Rightarrow u(t) = \sum_{i=1}^n y_i(t) x_i = \sum \xi_i e^{\lambda_i t} x_i$$

$$\Rightarrow u = \xi_1 e^{\lambda_1 t} x_1 + \xi_2 e^{\lambda_2 t} x_2 + \dots + \xi_n e^{\lambda_n t} x_n$$

for $u(0) = c \Rightarrow P\xi = c$, we can write:

$$u(t) = u(0) = \xi_1 e^{\lambda_1 t} x_1 + \xi_2 e^{\lambda_2 t} x_2 + \dots + \xi_n e^{\lambda_n t} x_n = \xi_1 x_1 = I\xi \Rightarrow P\xi = c, \xi = [\xi_1 | \dots | \xi_n]^T$$

$$\xi = P^{-1} c$$

$$\Rightarrow u(t) = \xi_1 e^{\lambda_1 t} x_1 + \xi_2 e^{\lambda_2 t} x_2 + \dots + \xi_n e^{\lambda_n t} x_n, \xi = P^{-1} c, \xi = (\xi_1, \dots, \xi_n)^T$$

5 m_1, m_2, m_3 a) $y_i(t)$ displacement, $M y'' = K y$,

$$M = \begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}, \quad y = \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}, \quad K = K \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$



$$\begin{cases} m_1 y_1'' = -K y_1 + K(y_2 - y_1) = -2K y_1 + K y_2 \\ m_2 y_2'' = -K(y_2 - y_1) + K(y_3 - y_2) = K y_1 - 2K y_2 + K y_3 \\ m_3 y_3'' = -K(y_3 - y_2) = K y_2 - K y_3 \end{cases} \Rightarrow \begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{bmatrix} y_1'' \\ y_2'' \\ y_3'' \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$\Rightarrow M y'' = K y$

b) K is positive definite15 For potential we have $\frac{1}{2} y^T K y$, $y^T K y > 0$, $y \neq 0$, then we can write:

$$y^T K y = y^T \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = K(y_1^2 + (y_1 - y_2)^2 + (y_2 - y_3)^2)$$

For $K > 0$, $y \neq 0 \Rightarrow y^T K y > 0$: K is positive definite

20

c) normal modes when $m_1 = m_2 = m_3$

$$\begin{aligned} m_1 = m_2 = m_3 = m &\Rightarrow M = mI \Rightarrow M y'' + K y = 0 \Rightarrow m y'' + K y = 0 \Rightarrow y'' = -\frac{K}{m} y \\ \text{displacement consider as: } y(t) = v e^{i\omega t} &\Rightarrow y' = -\omega^2 v e^{i\omega t} \Rightarrow y''(t) = -\omega^2 y(t) \end{aligned} \quad \left. \begin{array}{l} K = K A \\ y'' = -\frac{K}{m} y \end{array} \right\}$$

$$-\omega^2 v = -\frac{K}{m} A v \Rightarrow \omega v = A v, \quad \lambda = \frac{m}{K} \omega^2 \Rightarrow A v = \lambda v \Rightarrow \det(A - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 - 5\lambda^2 + 6\lambda - 1 = 0 \Rightarrow \lambda_1 = 0.19806, \lambda_2 = 1.55496, \lambda_3 = 3.24698$$

$$\lambda = \frac{m}{K} \omega^2 \Rightarrow \omega^2 = \frac{K}{m} \lambda \Rightarrow \omega = \left(\frac{K}{m} \lambda \right)^{1/2} \Rightarrow \omega_i = \left(\frac{K}{m} \lambda_i \right)^{1/2} \text{ for } i=1,2,3$$

for λ_i :

$$\begin{bmatrix} 2-\lambda_i & -1 & 0 \\ -1 & 2-\lambda_i & -1 \\ 0 & -1 & 1-\lambda_i \end{bmatrix} v = 0 \Rightarrow \begin{cases} x_1 = r \\ x_2 = (2-\lambda_i)x_1 = (2-\lambda_i)r \\ x_3 = \frac{1}{1-\lambda_i} x_2 = \frac{2-\lambda_i}{1-\lambda_i} r \end{cases} \Rightarrow v_i = \begin{bmatrix} 1 \\ 2-\lambda_i \\ \frac{2-\lambda_i}{1-\lambda_i} \end{bmatrix}$$

$$\Rightarrow v = \left[1 \quad 2-\lambda_i \quad \frac{2-\lambda_i}{1-\lambda_i} \right], \lambda = \lambda_i, i=1,2,3 : \text{normal modes}$$

$$\underline{6} \quad A_{n \times n} \Rightarrow \lambda_i \text{ distinct}, \Rightarrow P(A) = c_1 A^{n-1} + c_2 A^{n-2} + \dots + c_{n-1} A + c_n I$$

$$P(\lambda_i) = \sum_{k=1}^n c_k \lambda_i^{n-k}$$

A is symmetric with distinct eigenvalues, characteristic polynomial is:

$$m(t) = \prod_{i=1}^n (t - \lambda_i), \text{ degree of } n \text{ and root of eigenvalues.}$$

we can write scalar polynomial $q(t)$ divide by $m(t)$:

$$q(t) = s(t)m(t) + r(t) \Rightarrow \text{degree of } r(t) \text{ (remainder) if less than } \deg(m):$$

$$\deg r(t) < \deg m(t) = n \Rightarrow r(t) = c_1 t^{n-1} + c_2 t^{n-2} + \dots + c_{n-1} t + c_n$$

we can write: $q(A) = s(A)m(A) + r(A)$, from Cayley-Hamilton we know:

$$m(A) = 0 \Rightarrow q(A) = 0 + r(A) \Rightarrow q(A) = r(A) = c_1 A^{n-1} + c_2 A^{n-2} + \dots + c_{n-1} A + c_n I = P(A)$$

For uniquely, we consider have two representation:

$$P(A) = \sum_{k=1}^n c_k A^{n-k} = \sum_{k=1}^n d_k A^{n-k} \Rightarrow \sum_{k=1}^n (c_k - d_k) A^{n-k} = 0, \text{ write scalar polynomial:}$$

$$g(t) = \sum_{k=1}^n (c_k - d_k) t^{n-k}, g(A) = 0$$

$$\text{for diagonalizable } A \text{ with } \lambda_i, A x_i = \lambda_i x_i \Rightarrow g(A) x_i = g(\lambda_i) x_i$$

$$\text{we had: } g(A) = 0 \Rightarrow g(A) x_i = 0 \Rightarrow g(\lambda_i) x_i = 0 \Rightarrow \text{for } x_i \neq 0: g(\lambda_i) = 0$$

$$\text{For } g(\lambda_i) = 0 \Rightarrow \text{roots: } \lambda_1, \dots, \lambda_n \text{ distinct} \Rightarrow g(t) = 0 \Rightarrow c_k - d_k = 0 \Rightarrow c_k = d_k$$

$$\Rightarrow P(A) = \sum_{k=1}^n c_k A^{n-k} \text{ is unique}$$

$$\text{in } P(t) \text{ we plug } \lambda_i: P(t) = c_1 t^{n-1} + c_2 t^{n-2} + \dots + c_{n-1} t + c_n$$

$$= c_1 \lambda_i^{n-1} + c_2 \lambda_i^{n-2} + \dots + c_{n-1} \lambda_i + c_n = \sum_{k=1}^n c_k \lambda_i^{n-k}$$

for unknown C_k , have linear system:

$$\begin{cases} P(\lambda_1) = \sum C_k \lambda_1^{n-k} \\ P(\lambda_2) = \sum C_k \lambda_2^{n-k} = 0 \\ \dots \\ P(\lambda_n) = \sum C_k \lambda_n^{n-k} \end{cases} \Rightarrow \begin{pmatrix} \lambda_1^{n-1} & \lambda_1^{n-2} & \dots & 1 \\ \lambda_2^{n-1} & \lambda_2^{n-2} & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_n^{n-1} & \lambda_n^{n-2} & \dots & 1 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{pmatrix} = \begin{pmatrix} P(\lambda_1) \\ 0 \\ \vdots \\ P(\lambda_n) \end{pmatrix}$$

by simplification, can write eigenvalue vectors as: $\det(H) = \prod_{1 \leq i < j \leq n} (\lambda_j - \lambda_i)$

It is nonzero if just $\lambda_i = \lambda_j$ and we know eigen values are distinct:

$\lambda_i \neq \lambda_j \Rightarrow \lambda_i - \lambda_j \neq 0 \Rightarrow \det(H) \neq 0 \Rightarrow$ full rank and unique solution for C .
* invertible

10. $q = 3x_1^2 + 3x_2^2 + 3x_3^2 + 2x_1x_2 - 10x_1x_3 + 10x_1x_4 + 10x_2x_3 - 10x_2x_4 + 2x_3x_4$, diagonalize
 $Y = (y_1, y_2, y_3, y_4)^T$, $q = Y^T A Y$, $D = P^T A P$, $X = P Y$

from def. of q , we can write A matrix as:

15. $A = \begin{pmatrix} 3 & 1 & -5 & 5 \\ 1 & 3 & 5 & -5 \\ -5 & 5 & 3 & 1 \\ 5 & -5 & 1 & 0 \end{pmatrix}$ $C_{xixi} = C = a_{ii}$, $C_{xixj} = C = \frac{1}{2} a_{ij}$
 $\det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} 3-\lambda & 1 & -5 & 5 \\ 1 & 3-\lambda & 5 & -5 \\ -5 & 5 & 3-\lambda & 1 \\ 5 & -5 & 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 11.396, \lambda_2 = 4, \lambda_3 = 2.489, \lambda_4 = -8.885$

$\Rightarrow \lambda_1 = 11.4, \lambda_2 = 4, \lambda_3 = 2.5, \lambda_4 = -8.9$

20. For $\lambda_1 = 11.4$:

For $\lambda_2 = 4$:

$\begin{pmatrix} -8.4 & 1 & -5 & 5 \\ 1 & -8.4 & 5 & -5 \\ -5 & 5 & -8.4 & 1 \\ 5 & -5 & 1 & -11.4 \end{pmatrix} v_1 = 0 \Rightarrow \begin{pmatrix} -0.512 \\ 0.512 \\ 0.562 \\ -0.4 \end{pmatrix}$, $\begin{pmatrix} -1 & 1 & -5 & 5 \\ 1 & -1 & 5 & -5 \\ -5 & 5 & -1 & 1 \\ 5 & -5 & 1 & -4 \end{pmatrix} v_2 = 0 \Rightarrow \begin{pmatrix} -0.707 \\ 0.707 \\ 0 \\ 0 \end{pmatrix}$

$\lambda_3 = 2.5$

$\lambda = -8.9$

25. $\begin{pmatrix} 0.5 & 1 & -5 & 5 \\ 1 & 0.5 & 5 & -5 \\ -5 & 5 & 0.5 & 1 \\ 5 & -5 & 1 & -2.5 \end{pmatrix} v_3 = 0 \Rightarrow v_3 = \begin{pmatrix} 0.1059 \\ -0.1059 \\ 0.6989 \\ 0.7043 \end{pmatrix}$, $\begin{pmatrix} 11.9 & 1 & -5 & 5 \\ 1 & 11.9 & 5 & -5 \\ -5 & 5 & 11.9 & 1 \\ 5 & -5 & 1 & 8.9 \end{pmatrix} v_4 = 0 \Rightarrow v_4 = \begin{pmatrix} 0.4762 \\ -0.4762 \\ 0.4500 \\ -0.5866 \end{pmatrix}$

$$P = \begin{bmatrix} -0.512 & -0.707 & 0.1059 & 0.4762 \\ 0.512 & 0.707 & -0.1059 & -0.4762 \\ 0.562 & 0 & 0.6939 & 0.4500 \\ -0.4 & 0 & 0.7043 & -0.5866 \end{bmatrix} \quad P^{-1} = P^T = \begin{bmatrix} -0.512 & 0.512 & 0.562 & -0.4 \\ -0.707 & 0.707 & 0 & 0 \\ 0.1059 & -0.1059 & 0.6939 & 0.7043 \\ 0.4762 & -0.4762 & 0.45 & -0.5866 \end{bmatrix}$$

orthogonal
eigenvector

$$X = PY \quad Y = P^{-1}X = P^T X = \begin{bmatrix} -0.512 & 0.512 & 0.562 & -0.4 \\ -0.707 & 0.707 & 0 & 0 \\ 0.1059 & -0.1059 & 0.6939 & 0.7043 \\ 0.4762 & -0.4762 & 0.45 & -0.5866 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$Y = \begin{bmatrix} -0.512x_1 + 0.512x_2 + 0.562x_3 - 0.4x_4 \\ -0.707x_1 + 0.707x_2 \\ 0.1059x_1 - 0.1059x_2 + 0.6939x_3 + 0.7043x_4 \\ 0.4762x_1 - 0.4762x_2 + 0.45x_3 - 0.5866x_4 \end{bmatrix}^T$$

$$8 \quad \begin{cases} y'' + \lambda^2 y = x^2 \\ y(0) - y'(0) = 0 \\ y(1) - y'(1) = 0 \end{cases} \quad \text{eigen function and eigenvalues}$$

Sturm-Liouville homogeneous problem:

$$y'' + \lambda^2 y = 0 \quad \text{for } \lambda \neq 0: y'' + \lambda^2 y = 0 \quad y = A \cos \lambda x + B \sin \lambda x, y' = -A \lambda \sin(\lambda x) + B \lambda \cos(\lambda x)$$

$$BC: y(0) - y'(0) = 0 \quad A - \lambda B = 0 \quad A = \lambda B$$

$$y(1) - y'(1) = 0 \quad y(1) = A \cos \lambda + B \sin \lambda = B(\lambda \cos \lambda + \sin \lambda)$$

$$y'(1) = -A \lambda \sin \lambda + B \lambda \cos \lambda = B(\lambda^2 \sin \lambda + \lambda \cos \lambda)$$

$$= B(\lambda \cos \lambda + \sin \lambda) - B(\lambda^2 \sin \lambda + \lambda \cos \lambda) = 0 \quad B \neq 0 \quad (1 + \lambda^2) \sin \lambda = 0 \quad \sin \lambda = 0 \quad \lambda = n\pi$$

$$\Rightarrow \lambda = n\pi \quad n = 1, 2, 3, \dots$$

$$\Rightarrow y_n(x) = A \cos(n\pi x) + B \sin(n\pi x) = B(n\pi \cos(n\pi x) + \sin(n\pi x)) \quad B=1$$

$$y_n = n\pi \cos(n\pi x) + \sin(n\pi x) \quad \text{eigenfunction, } \lambda_n = n\pi \quad \text{eigenvalues}$$

inhomogeneous equation:

$$y'' + \lambda^2 y = x^2 \quad y(x) = y_h(x) + y_p(x) \quad y_h = A \cos(\lambda x) + B \sin(\lambda x)$$

$$\text{polynomial: } y_p(x) = ax^2 + bx + c \quad y_p'' = 2a \quad 2a + \lambda^2(ax^2 + bx + c) = x^2$$

$$\lambda^2 a = 1 \quad a = 1/\lambda^2, \quad \lambda^2 b = 0 \quad b = 0, \quad \lambda^2 c + 2a = 0 \quad c = -2/\lambda^4 \quad y_p = \frac{x^2}{\lambda^2} - \frac{2}{\lambda^4}$$

$$\Rightarrow y(x) = A \cos(\lambda x) + B \sin(\lambda x) + \frac{x^2}{\lambda^2} - \frac{2}{\lambda^4}$$

$$BC: x=0 \Rightarrow y(0) - y'(0) = 0 \Rightarrow A - \frac{2}{\lambda^4} - \lambda B = 0$$

$$\lambda=1 \Rightarrow y(1) - y'(1) = 0 \Rightarrow A(\cos \lambda + \lambda \sin \lambda) + B(\sin \lambda - \lambda \cos \lambda) + \frac{1}{\lambda^2} - \frac{2}{\lambda^4} - \frac{2}{\lambda^2} = 0$$

$$\Rightarrow A = \lambda B + \frac{2}{\lambda^4}, \quad B = \frac{\lambda^2 + 2 - 2(\cos \lambda + \lambda \sin \lambda)}{\lambda^4(1 + \lambda^2) \sin \lambda}$$

5. general solution:

$$\Rightarrow y(x) = A \cos(\lambda x) + B \sin(\lambda x) + \frac{x^2}{\lambda^2} - \frac{2}{\lambda^4}, \quad A = \lambda B + \frac{2}{\lambda^4}, \quad B = \frac{\lambda^2 + 2 - 2(\cos \lambda + \lambda \sin \lambda)}{\lambda^4(1 + \lambda^2) \sin \lambda}$$

$$\text{eigenvalues: } \lambda n = n\pi, \quad n=1, 2, \dots, \quad \text{eigenfunction: } y_n(x) = n\pi (\cos(n\pi x) + \sin(n\pi x))$$

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